

Semantic-Aware Clustering-based Approach of Trajectory Data Stream Mining

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Abstract—The rapid development of mobile sensing technologies (like GPS, RFID, accelerometer, gyroscope etc. various sensors in smart phones) has caused a rise in the large-scale capture of positioning data. These mobility data generated by heterogeneous mobile devices with embedded sensors are mostly known as trajectories. There are different algorithms to process the large amounts of mobility data for identifying mobility patterns. Even though few of these algorithms consider the use of semantic annotations on the data, none of the existing research has considered semantic annotations for online sub-trajectory clustering-based movement behavior analysis. In this paper, we incorporate semantics annotation in the raw trajectory data in order to discover various movement relationships between sub-trajectories of mobile devices. We conduct experiment on a real-world data set. Along with the added advantage of semantic-aware movement behavior analysis, our method is able to identify outliers in the clustering process with almost similar performance (average recall 0.92) as classic density-based clustering algorithm DBSCAN.

I. INTRODUCTION

Due to the growth of various mobile devices, there is also growth in the mobile trajectories. As a result, there comes the research opportunity to find the movement relationship between these mobile nodes. In some relationships, only one entity is being considered; in others inter trajectory relationships are considered. For example, stop and go behavior can be found in a single trajectory. Moving together, flock, leader - follower are some of the group movement behaviors. There exist some efforts on using the whole trajectory in order to find clusters of mobile nodes [10], [16].

We need to deal with enormous amount of trajectory data generated by the mobile devices that change their positions frequently to discover various movement relationships. In order to analyze their movement behavior, offline methods are not suitable. The consideration of the entire data stream might not be very useful, because the information to be extracted may be outdated at the time of processing [5]. To handle this issue, we develop an online method that performs operation on the input data streams in real-time rather than traditional offline trajectory clustering and analysis methods. By looking at the recent time windows, we apply clustering on sub-trajectories to discover similarity in movement behaviors as well as identify outliers in real-time. Outliers have been

defined as observations inconsistent with the remainder of that set of data [4]. The identification of outliers can lead to the discovery of useful and meaningful knowledge and has a number of practical applications (e.g., transportation, location based services, public safety etc.).

Semantics is the piece of information that allows us to give a meaningful value to the data; for example, GPS coordinates consist of latitude and longitude, the semantic annotation will give those points a human readable label such as “restaurant,” “school,” “bank” etc. Adding semantic annotation facilitates various movement and behavioral pattern in mobile trajectory data. In this paper, we propose a Semantic-Aware Clustering-based (SACB) approach for trajectory data stream mining. We use DBSCAN [6] as our baseline algorithm.

The main contributions of this paper are as follows.

- We propose a method of adding geographic domain knowledge to raw trajectory data
- We propose a window-based online sub-trajectory clustering method for finding movement similarity based on space, time, direction and semantics
- We implement the SACB on a real-world application dataset that contains mobility traces of taxi cabs in San Francisco, USA [14]

We organize our paper as following. In section II, we discuss our work with related work. In section III, we discuss different modules of our overall system and present algorithm for stop point identification and semantic-aware trajectory clustering. In section IV, we describe the experimental setup, and discuss the evaluation results. Finally, concluding remarks and future works are offered in section V.

II. RELATED WORK

In this section, we discuss the works that are most pertinent to our work. Data mining is a crucial part of data management. It is the process used by the user to discover meaningful information in vast data repositories. Appiece et al. [3] presents three most popular data mining techniques: predictive modeling, clustering analysis and anomaly analysis. Data management for wireless sensor networks has been a topic of interest. Data management covers everything from storage, compression, prediction, optimization and even data

cleaning. As of today, there are a wide range of algorithms that deal with data management. Jeffrey et al. [7] explains why data cleaning is a crucial step for data management. Samia et al. [15] presented a context-aware model for cleaning data stream. Zafeiropoulus et al. [21] went a step further and discussed the advantages of adding a semantic component to the data, how those semantic annotations enrich the data and ultimately allow for more specific queries. As an example, a museum worker would be able to search for the location of the paintings of a specific artist or a specific time period.

Zhixian Yan et al. [17] presented a multi-stage model to cope with the complexity of trajectory semantics. The multi-stage model takes raw movement data, and creates raw trajectories which are later converted to structured trajectories. Mingqi et al. [9] presents a framework that discovers the users semantic places from their GPS data. This framework includes a multi-layered model that extracts physical places by using a density-based algorithm.

There are some works on the clustering of mobile objects based on the whole trajectory of the movement nodes [10], [13], [16]. In these cases, the whole lifetime of the trajectories is looked into for discovering cluster. In [8], the authors considered sub-trajectory based clustering of trajectories considering only spatial relationship. However, since the inclusion of time was missing, it is not capable of finding out the exact relationship between various mobile trajectories. Spatial clustering is used for moving object activity discovery. Silva et al. [5] used sub-trajectory based clustering considering space and time. Though, they did not include semantics in their trajectory clustering method. Palma et al. [11] used a density-based clustering algorithm to find interesting places from trajectory. There is also research on shape based clustering that look into the shape of the trajectory for similarity finding. A shape-based similarity query is defined in [18]. In contrast, there is significant work based on time dependent clustering. In this type of works, periodic patterns among movement trajectories are analyzed [10], [19]. Ying et al. [20] proposed a novel predict model based on cluster-based prediction strategy to find out the next location of a mobile user based on frequent behavior of similar users in semantic trajectory.

However, to the best of our knowledge, no work has been done on online sub-trajectory clustering-based movement behavior analysis on semantic annotated trajectory data stream.

III. METHODOLOGY

A. Add Geographic Domain Knowledge

In the raw data as shown in Table I, only latitude, longitude values of the movement points are available. This module takes care of adding geographic domain knowledge to the points. As a result, the point receives a human understandable format. We add the type of the place that the mobile object visited using this module. We consider the places that are in 15 m radius (which is common sensing range) of the trajectory points. Table II presents the addition of geographic domain knowledge on the raw data as shown in Table I. We used

the reverse GeoCoding API [1] to get places within a pre-determined radius.

TABLE I
SAMPLE RAW DATA

Latitude	Longitude
37.78574	-122.4146
37.64401	-122.45264
37.79434	-122.39983

TABLE II
SAMPLE DATA AFTER ADDITION OF GEOGRAPHIC INFORMATION

Place Type	Latitude	Longitude
atm	37.78574	-122.4146
restaurant	37.64401	-122.45264
bank	37.79434	-122.39983

B. Distance Function

In this section, we measure the distance between sub-trajectories that are concurrently present in the considered time window ($t, t + |window|$). We map the trajectory points (latitude, longitude) into euclidean space in order to measure the distance between sub-trajectories in spatio-temporal dimension. If the sampling rates of a pair of sub-trajectories vary, we used linear interpolation to approximate the missing location point. The Euclidean distance between two objects that move with in same time window is calculated using eqn. 1.

$$\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} \quad (1)$$

C. Direction Calculate

Forward azimuth is used to calculate the direction of movement for the trajectory points [2]. Based on the calculated angle, the direction is defined. Different directions are : E, N, S, W, NE, NW, SE, SW, where E is for the EAST direction, N for the NORTH, S for SOUTH, W is for WEST, NE is for NORTH-EAST, NW is for NORTH-WEST, SE is for SOUTH-EAST, and SW is for SOUTH-WEST. The Cosine function is applied on the angle to calculate the directional distance.

D. Stop and Move Annotation

We find episodes in the trajectory. Based on the movement speed and duration of staying at a particular point, Algorithm 2 finds out the stop points in the trajectory. We have used a clustering-based method in order to annotate stops and moves in the trajectory. If the speed of a certain number of trajectory points is less than a threshold for a certain duration, then it is considered as stop.

Algorithm 1 CreateClusters**Input :** Trajectories, range

```

1: ListOfClusters = []
2: for all point  $p \in \text{Trajectory}$  do
3:   for all cluster  $x \in \text{ListOfClusters}$  do
4:     if Latitude in range & Longitude in range then
5:       ListOfClusters[ $x$ ].append( $p$ )
6:     else
7:       ListOfClusters.newCluster( $p$ )
8:     end if
9:   end for
10: end for
11: return ListOfClusters[]

```

Algorithm 2 Find Stops**Input :** Trajectory

```

1: ListOfClusters = CreateClusters(Trajectories, rng)
2: Stops = []
3: for all point  $p \in \text{Trajectory}$  do
4:   for all cluster  $x \in \text{ListOfClusters}$  do
5:     if time in range & speed in range then
6:       stops[ $x$ ].append( $p$ )
7:     else
8:       moves[ $x$ ].append( $p$ )
9:     end if
10:   end for
11: end for
12: return stops[], moves[]

```

E. Group Formation

This module considers the semantic relationship between the mobile nodes. We follow a window-based mechanism in order to find out the group of trajectories that have similar behavior. There are few steps that are maintained in order to achieve this. Such as :

- (i) collect the trajectory data stream at each time window
- (ii) apply a similarity measure
- (iii) maintain the group(s)
- (iv) discover the mobility

Algorithm 3 describes the formation of group of trajectories that are similar based on time, space, direction and semantics. We use string matching in order to find semantic distance. It is a window based density clustering algorithm where we look at the sub trajectories of all moving trajectories that are present during that time period.

We use linear interpolation to align the points in time in order to measure the distance between a pair. The three weights for calculating distance has equal weight value in Eqn. 2 . If for some particular scenario spatio-temporal distance needs more emphasis than direction, then the weight value w_1 will receive higher value. These three weight values are application dependent.

Algorithm 3 Group Formation**Input :** S_i sub-trajectory set for the time $window_i = [t, t + |window_i|]$, α distance threshold, ω the size threshold, ρ representativeness threshold

```

1:  $G_i \leftarrow \text{empty}$ 
2: mark all the sub-trajectories in  $S_i$  as unvisited
3: while  $\exists (obj_j, SubT_{j,i}) \in S_i$  unvisited do
4:   mark( $obj_j, SubT_{j,i}$ ) as visited
5:   get neighborlist  $N_\alpha(obj_j)$ 
6:   if  $|N_\alpha(obj_j) - 1| \geq \omega$  then
7:     for all  $obj_k \in N_\alpha(obj_j)$  not visited do
8:        $dist \leftarrow \text{Equation (2)}$ 
9:       if  $e^{-dist^2 \div 2\sigma^2} > \rho$  then
10:         $g \leftarrow g \cup \{o_k\}$ 
11:        mark ( $obj_k, SubT_{k,i}$ ) as visited
12:      end if
13:    end for
14:     $representative \leftarrow (obj_j, SubT_{j,i})$ 
15:     $G_i \leftarrow G_i \cup \{g\}$ 
16:  else
17:    mark  $SubT_{j,i}$  as ungrouped
18:  end if
19: end while
20: for all  $SubT_{j,i}$  ungrouped do
21:   mark  $obj_j$  as outlier
22: end for
23: return  $G_j$ 

```

$$\begin{aligned}
dist(SubT_{j,i}, SubT_{k,i}) = & w_1 * distance(SubT_{j,i}, SubT_{k,i}) + \\
& w_2 * direction(SubT_{j,i}, SubT_{k,i}) + \\
& w_3 * semantic(SubT_{j,i}, SubT_{k,i}) \quad (2)
\end{aligned}$$

Now, we briefly explain the Algorithm 3. First, we select a sub-trajectory of a moving object ($obj_j, SubT_{j,i}$) that is unvisited. It is a random selection. Then, we mark this sub-trajectory as visited and check if it can be a representative trajectory. The representative trajectory is identified following gaussian kernel based voting method [12]. If it has minimum $\omega - 1$ neighbors in α radius, then a new group “g” is created including itself and the neighbors that are represented by ($obj_j, SubT_{j,i}$). Else, obj_j is marked as ungrouped, because it cannot be represented by another representative that belongs to an existing group or it can be an outlier. At the end of the while loop, the ungrouped objects are declared as outliers. The Algorithm 3 outputs a set of groups. By applying a post processing mechanism on the groups, the leader among the group members is detected.

IV. PERFORMANCE EVALUATION

In this section we describe our experimental setup and environment. We also describe the data set we used and different performance measures that we considered to evaluate the accuracy of our proposed method. For implementing our

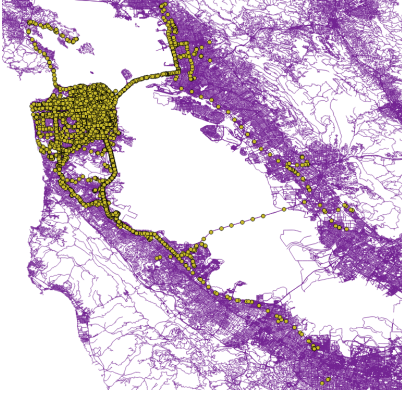


Fig. 1. Taxi trajectory points on the map of San Francisco



Fig. 2. Stop points identified from all the taxi trajectories

algorithm, we have used the real dataset [14]. The dataset consists of gps positions of taxi cabs that move around the city of San Francisco. The area of consideration has the size of 42.5 km X 62.5 km. There is in total 160 taxis that were moving around different parts of the city. Fig. 1 depicts the raw point of movement taxis.

After clustering-based stop and go Algorithm 2, we get the stop and go episodes. We used the reverse GeoCoding API [1] to get places within a pre-determined radius. After adding geographic information to the raw gps points, we get the semantic annotated trajectory. These semantic annotated trajectories are used as input in Algorithm 2 and Algorithm 3. As a result, we can find stop episodes and group formation among trajectories respectively. The stop points identified from all the taxi trajectories are shown on the map in Fig. 2.

Due to adding semantic information, we could classify the data based on weekend and weekdays. We analyzed the semantics of the stop points and the categorization on weekend is shown in Fig. 3.

$$recall = \frac{|SACB_outlier \cap DBSCAN_outlier|}{|SACB_outlier|} \quad (3)$$

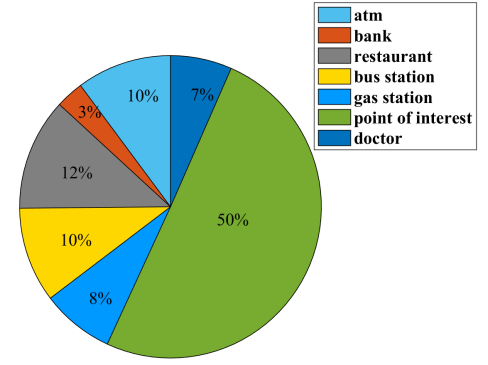


Fig. 3. Stop points identified from all the taxi trajectories

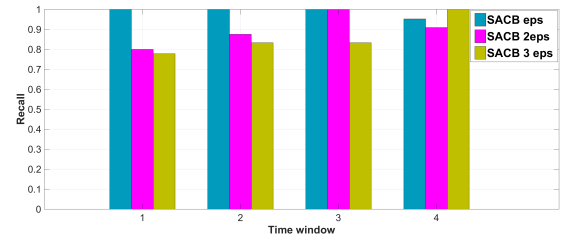


Fig. 4. Recall for detecting outlier

$$precision = \frac{|SACB_outlier \cap DBSCAN_outlier|}{|DBSCAN_outlier|} \quad (4)$$

After applying Algorithm 3, we identified the cluster of taxis as well as outlier taxis. The taxi that does not belong to any group is declared as outlier. We captured timestamps from 9 : 42 : 00 to 10 : 22 : 09 in May 17, 2008 as the experimental dataset to be used in implementing Algorithm 3. We set the time window as 5 minutes and tracked the clusters for 4 consecutive time windows. In Fig. 4, we show the performance of our algorithm in detecting outlier taxis. The recall values achieved during outlier identification at various time windows are shown. Three different clustering radius (*eps*, *2eps*, *3eps*) has been tested. We used 250 as the *eps* value. Here, we consider the outliers identified by DBSCAN as the perfect list. The recall and precision is calculated using eqn. 3 and eqn.4. *SACB_outlier* is the list of taxis identified as outlier by our algorithm and *DBSCAN_outlier* contains the outlier taxis identified by the widely used density-based clustering algorithm : DBSCAN.

We tested with different density requirement (ω) and representativeness threshold (ρ) values. In Fig. 5, we show recall values with respect to different node density as well as representativeness threshold value. The range of ρ is between 0 and 1.0. $\rho = 1$ means the highest similarity between a pair of objects, the best candidate to be declared as representative among the group members and later leader in the moving

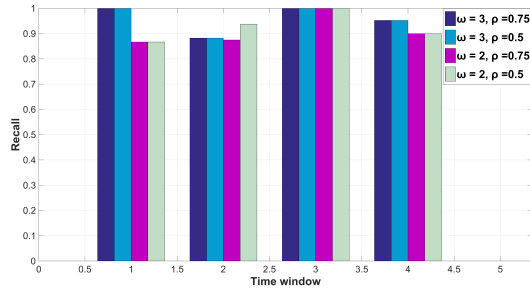


Fig. 5. Recall for detecting outlier

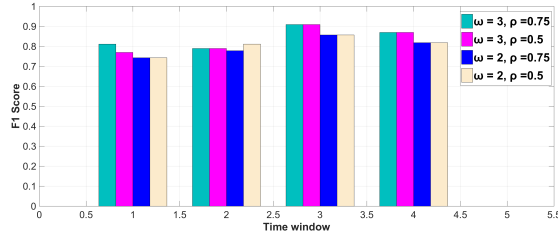


Fig. 6. F1 score for detecting outlier

group. Even in high ρ value, which demands close correspondence among the trajectories based on their spatio-temporal, directional and semantic relationship, our method was able to detect outlier taxis almost similar manner as DBSCAN by maintaining recall value over 0.85. The average recall value achieved was 0.92 for different combination of ω , α and ρ .

Fig. 6 shows the F1 score achieved at time window 1 – 4. F1 score is calculated using eqn. 5. The average F1 score encountered throughout the experiments was 0.8, which indicates the classification accuracy of our *SACB* method.

$$F1 = 2 * \frac{Precision * Recall}{Precision + Recall} \quad (5)$$

V. CONCLUSION & FUTURE WORK

In this paper, we introduced a semantic-aware online clustering-based method for movement relationships finding in mobile trajectories. We used semantic annotation to annotate the movement trajectories. After adding the geographic information to raw gps points, window-based online clustering is applied to find sub-trajectory groups. Ours is an online method that performs operation on the input streams in real-time rather than traditional offline trajectory clustering and analysis methods. We conducted experiment on a real-world data set. Along with the added advantage of semantic-aware movement behavior analysis, our method was able to successfully identify outliers in the clustering process with average recall 0.92 and F1 score of 0.8. In the future, we want to add different application domain knowledge in the mobile trajectory data. Furthermore, we want to find out additional type of movement patterns in the future.

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